Digital Logic System

Assignment 3

1. What are the steps required to implement any combinational circuit with NAND gate only?
2. What are the steps required to implement any combinational circuit with NOR gate only?
3. How would you design excess-3 to BCD code converter?
4. Obtain a full adder circuit using two half adders and one OR gate.
5. Design a combinational circuit with 3 inputs and 3 output. When the input is 0,1,2 and 3 the output is 1 more and when the input is 4,5,6 and 7 the output is 1 less than input.
6. Design a combinational circuit that converts a decimal digit from 2,4,2,1 code to 8,4,-2,-1 code.
7. What are the differences between multiplexer and demultiplexer?
8. Design half adder, full adder, half subtractor and full subtractor.
9. What do you mean by encoder and decoder? Explain the working principle of a decoder with diagram.
10. Design a combinational circuit that has four inputs and one output. The output is high only when the majority of inputs are high.
11. Design a combinational logic circuit that accepts a three bit number and generates an output binary number equal to the square of the input number.
12. Define literal. Design a circuit for the following Boolean function using only NOR gate. $F = xy'z + xyz' + (x' + y)z$
13. Design a BCD adder.
14. Design a parity generator and parity checker.
15. Write short notes on ROM and PLA.